Presented by Manavhela Tondani

VIEWERSHIP CASE STUDY

Q1 2016 | South Africa DStv Audience Intelligence

10,989

TOTAL SESSIONS

5,375

UNIQUE VIEWERS

1,523 hrs

TOTAL VIEWING TIME

9.1 min

AVG SESSION LENGTH

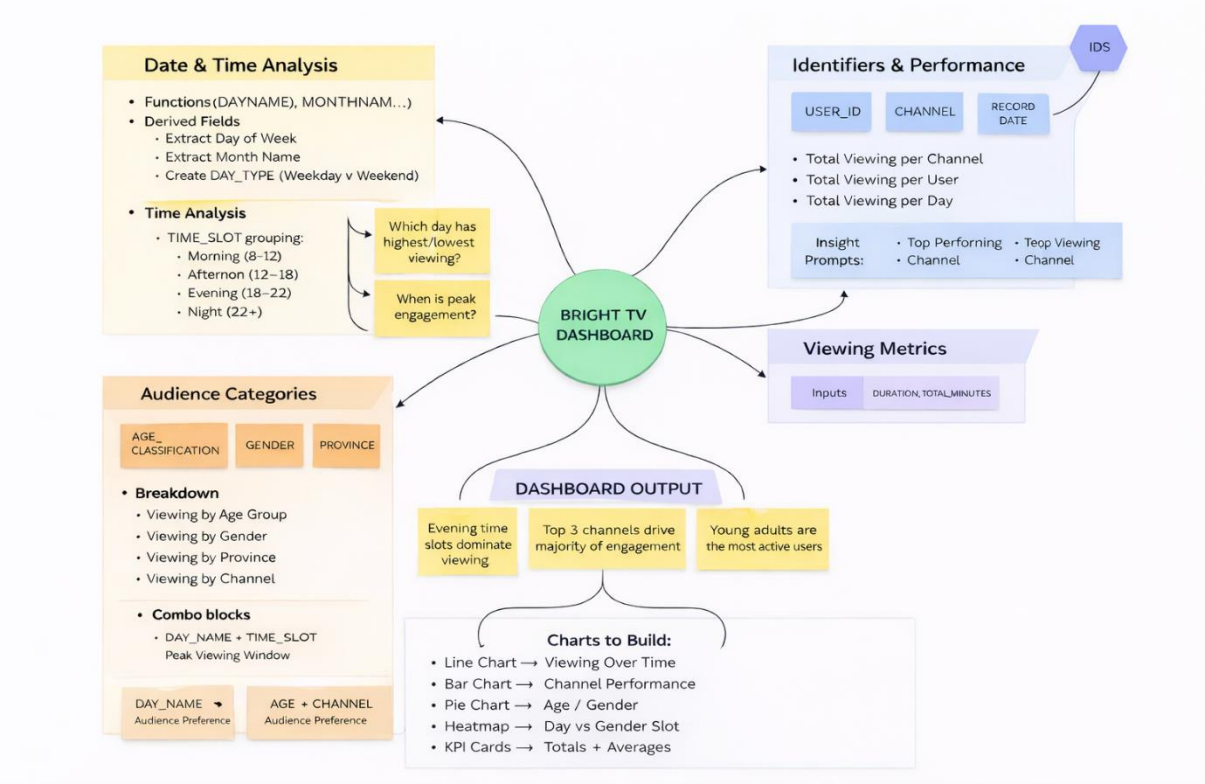
Jan – Apr 2016

1. Understanding the project and data

To further understand the project further a project plan, timeline and Gantt chart were developed to give out a breakdown of how the project will go.

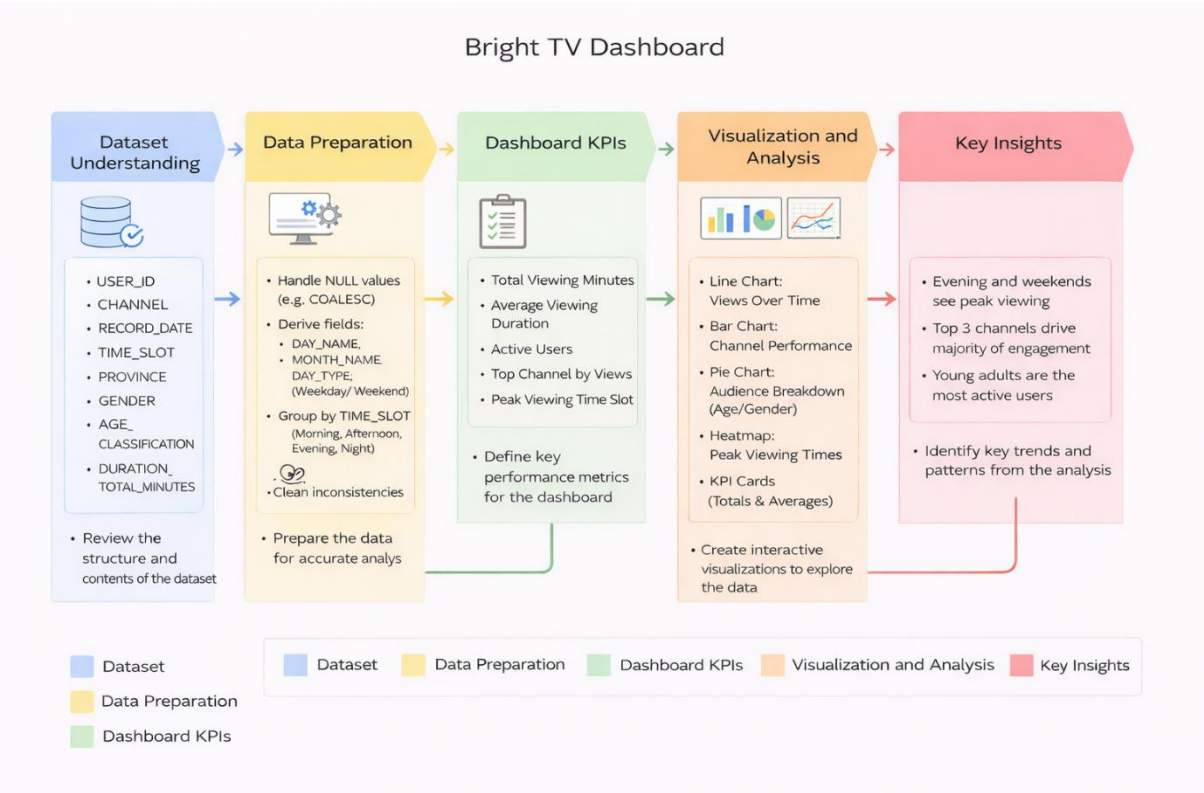
Project plan:

The project plan below outlines what will be included in the final project presentation, for this project it was vital to ensure that Date & Time analysis was done, Identifiers & performance together with understanding the audience categories.



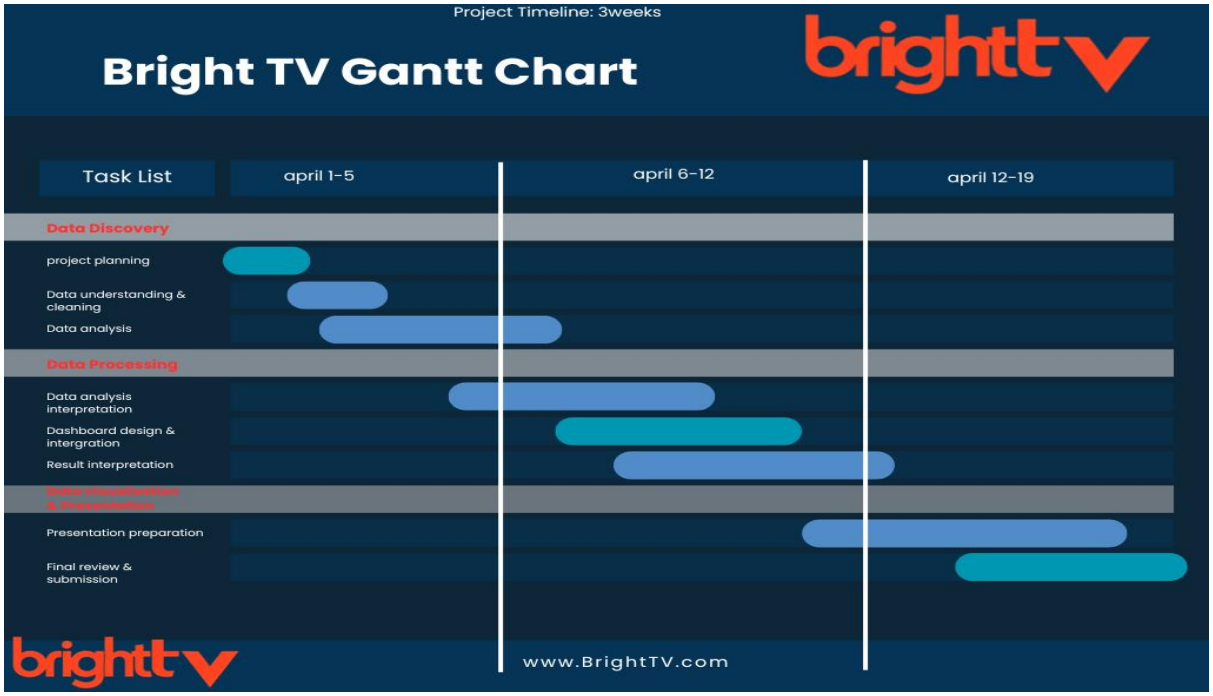
Project plan timeline:

The project plan timeline started with the understanding of datasets, then preparing it for presentation through data bricks, using SQL. The results were the used to develop KPI's which are then used for visualization and deriving key insights for the project.



Project Gannt chart:

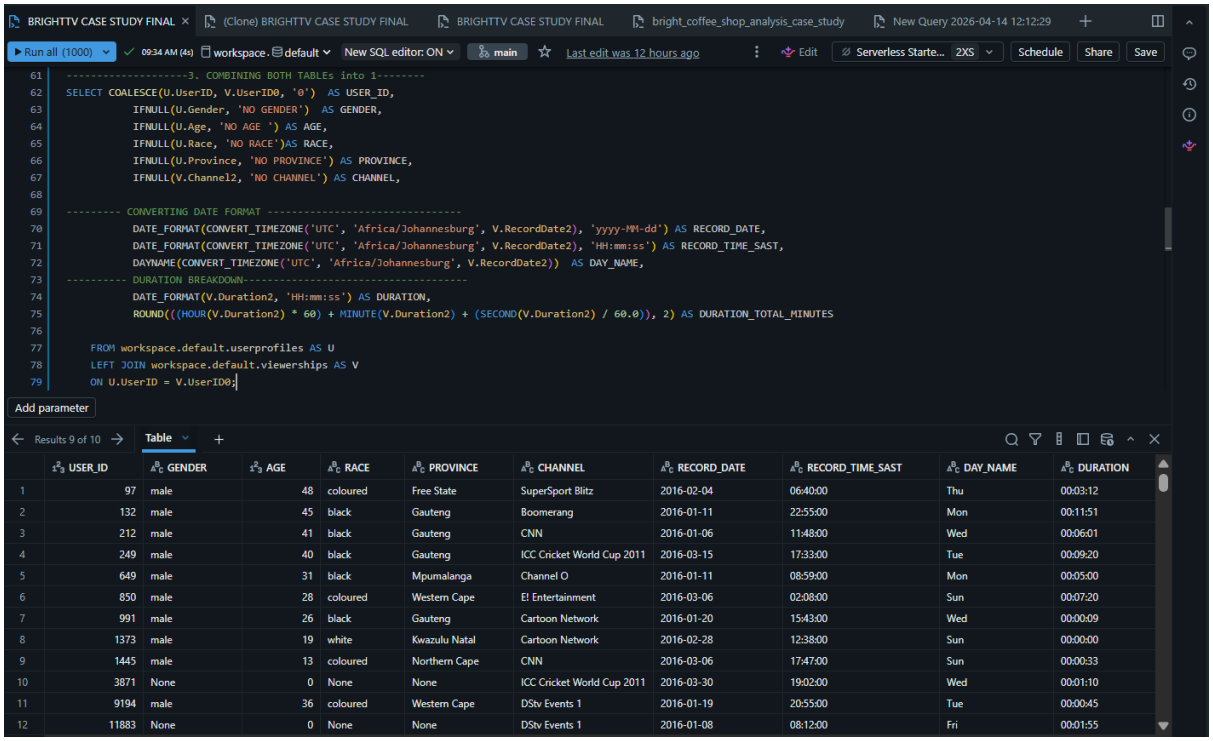
The project Gannt chart was only used to keep track of time, documentation and accountability.



2. SQL querying

Combining User profile viewership table

There are 2 separate tables inside one excels document, the best option was to divide the two tables separately and use a left join to join them together. Also in the excel document only the important titles such as userID, channel, month, etc… will only be extracted and not everything. Below is a databricks SQL query example. While joining them using a left joint, it is also important to convert the time utc to south African utc time.



Interpreting and understanding data

Started by extracting the whole table, then understanding the amount of people that watch TV, to which the findings reveal that the amount of people that watch Tv are 5375.

```sql
----------1. Extracting and  sorting the userprofile table-----

SELECT *
FROM workspace.default.userprofiles
LIMIT 1000;
----------1.1. COUNTING THE NUMBER OF USERS
SELECT COUNT(DISTINCT UserID ) AS NO_OF_VIEWERS
FROM workspace.default.userprofiles;
```

dd parameter

Results 2 of 10 → Table +

NO_OF_VIEWERS
5375

Filtering and removing NULLS and errors.

I used NULL function to filter all the data from any null that might exist as a way to make sure that all the data is filtered well and can be interpreted and presented.

```sql
-------- 2. Extracting and  sorting the viewership table......

SELECT *
FROM workspace.default.viewerships
LIMIT 1000;
-------------2.1 FILTERING AND SORTING VIEWWERSHIP TABLE----------
SELECT DISTINCT
        IFNULL(UserID0, '0')  AS USERID,
        IFNULL(Channel2, 'NO CHANNEL') AS CHANNELS,
        IFNULL(Duration2, 'NO DURATION') AS DURATION,
        IFNULL(RecordDate2, 'NO RECORD DATE') AS RECORD_DATE,
        YEAR(RecordDate2) AS YEAR_RECORDED,
        MONTH(RecordDate2) AS MONTH_RECORDED,
        DAYNAME(RecordDate2) AS DAY_RECORDED
FROM workspace.default.viewerships;
-------------------------------------
```

parameter

esults 6 of 10 → Table +

UserID0	Channel2	RecordDate2	Duration2	userid4
903994	Trace TV	2016-01-04T18:09:00.000+00:...	1899-12-31T00:02:00.000+00:...	903994
802352	ICC Cricket World Cup 2011	2016-03-30T09:04:00.000+00:...	1899-12-31T00:10:06.000+00:...	802352
866125	ICC Cricket World Cup 2011	2016-03-30T17:01:00.000+00:...	1899-12-31T00:00:42.000+00:...	866125
2425604	ICC Cricket World Cup 2011	2016-03-25T15:23:00.000+00:...	1899-12-31T00:00:10.000+00:...	2384970
793571	SawSee	2016-02-02T17:18:00.000+00:...	1899-12-31T00:01:33.000+00:...	793571
793571	Boomerang	2016-03-18T16:56:00.000+00:...	1899-12-31T00:00:10.000+00:...	793571
793571	Channel O	2016-03-20T04:25:00.000+00:...	1899-12-31T00:17:15.000+00:...	793571
793571	Boomerang	2016-02-03T06:43:00.000+00:...	1899-12-31T00:01:13.000+00:...	793571
819248	Channel O	2016-03-31T10:01:00.000+00:...	1899-12-31T00:04:45.000+00:...	819248
820321	Cartoon Network	2016-01-09T08:25:00.000+00:...	1899-12-31T00:00:40.000+00:...	820321
892453	Supersport Live Events	2016-03-24T09:09:00.000+00:...	1899-12-31T00:00:06.000+00:...	892453
892453	ICC Cricket World Cup 2011	2016-03-26T15:33:00.000+00:...	1899-12-31T05:01:12.000+00:...	892453
828799	Channel O	2016-02-28T08:09:00.000+00:...	1899-12-31T00:00:00.000+00:...	822560

1,000 rows | 4.30s runtime See performance (10) Refreshed 12 hours a

Using both tables to further interpret data.

To join both tables, I used the coalesce function to join all both tables and filter out all the nulls. Where I further made the joined tables a base as a way of keeping/making it a new table.

```sql
--------4. USING BOTH TABLES AS BASE TO ANALYSE FURTHER---------
WITH BASE AS (

    SELECT COALESCE(CAST(U.UserID AS STRING), CAST(V.UserID0 AS STRING), '0')                    AS USER_ID,
           IFNULL(U.Gender, 'NO GENDER')                                                          AS GENDER,
           IFNULL(U.Age, 0)                                                                       AS AGE,
           IFNULL(U.Race, 'NO RACE')                                                              AS RACE,
           IFNULL(U.Province, 'NO PROVINCE')                                                      AS PROVINCE,
           IFNULL(V.Channel2, 'NO CHANNEL')                                                       AS CHANNEL,

           -- CONVERTING DATE FORMAT
           DATE_FORMAT(CONVERT_TIMEZONE('UTC', 'Africa/Johannesburg', V.RecordDate2), 'yyyy-MM-dd') AS RECORD_DATE,
           DATE_FORMAT(CONVERT_TIMEZONE('UTC', 'Africa/Johannesburg', V.RecordDate2), 'HH:mm:ss')   AS RECORD_TIME_SAST,
           MONTHNAME(CONVERT_TIMEZONE('UTC', 'Africa/Johannesburg', V.RecordDate2))                AS MONTH_NAME,
           DAYNAME(CONVERT_TIMEZONE('UTC', 'Africa/Johannesburg', V.RecordDate2))                  AS DAY_NAME,

           -- DURATION BREAKDOWN
           DATE_FORMAT(V.Duration2, 'HH:mm:ss')                                                    AS DURATION,
           ROUND(((HOUR(V.Duration2) * 60) + MINUTE(V.Duration2) + (SECOND(V.Duration2) / 60.0)), 2) AS DURATION_TOTAL_MINUTES

    FROM workspace.default.userprofiles AS U
    LEFT JOIN workspace.default.viewerships AS V
    ON CAST(U.UserID AS STRING) = CAST(V.UserID0 AS STRING)

)
```

After making the base I further analysed to find the USER_ID, GENDER, AGE, AGE_CLASSIFICATION, RACE, PROVINCE, CHANNEL RECORD_DATE, MONTH_NAME, RECORD_TIME_SAST, TIME_SLOT, DAY_NAME, DURATION and DURATION_TOTAL_MINUTES so that I can create dashboards. Below is a detailed sql query description.

```sql
SELECT USER_ID,
       GENDER,
       AGE,
       CASE
           WHEN AGE = 0            THEN 'UNKNOWN'
           WHEN AGE < 13           THEN 'CHILD'
           WHEN AGE >= 13 AND AGE < 18 THEN 'ADOLESCENT TEEN'
           WHEN AGE >= 18 AND AGE < 25 THEN 'YOUNG ADULT'
           WHEN AGE >= 25 AND AGE < 45 THEN 'ADULT'
           WHEN AGE >= 45 AND AGE < 65 THEN 'MIDDLE-AGED ADULT'
           WHEN AGE >= 65          THEN 'SENIOR CITIZEN'
           ELSE 'OTHER'
       END  AS AGE_CLASSIFICATION,
       RACE,
       PROVINCE,
       CHANNEL,
       RECORD_DATE,
       MONTH_NAME,
       RECORD_TIME_SAST,
       CASE
           WHEN RECORD_TIME_SAST BETWEEN '05:00:00' AND '11:59:59' THEN 'MORNING SLOT'
           WHEN RECORD_TIME_SAST BETWEEN '12:00:00' AND '17:59:59' THEN 'AFTERNOON SLOT'
           WHEN RECORD_TIME_SAST >= '18:00:00'                     THEN 'EVENING SLOT'
           WHEN RECORD_TIME_SAST < '05:00:00'                      THEN 'MIDNIGHT OWL'
           ELSE 'MIDNIGHT OWL'
```

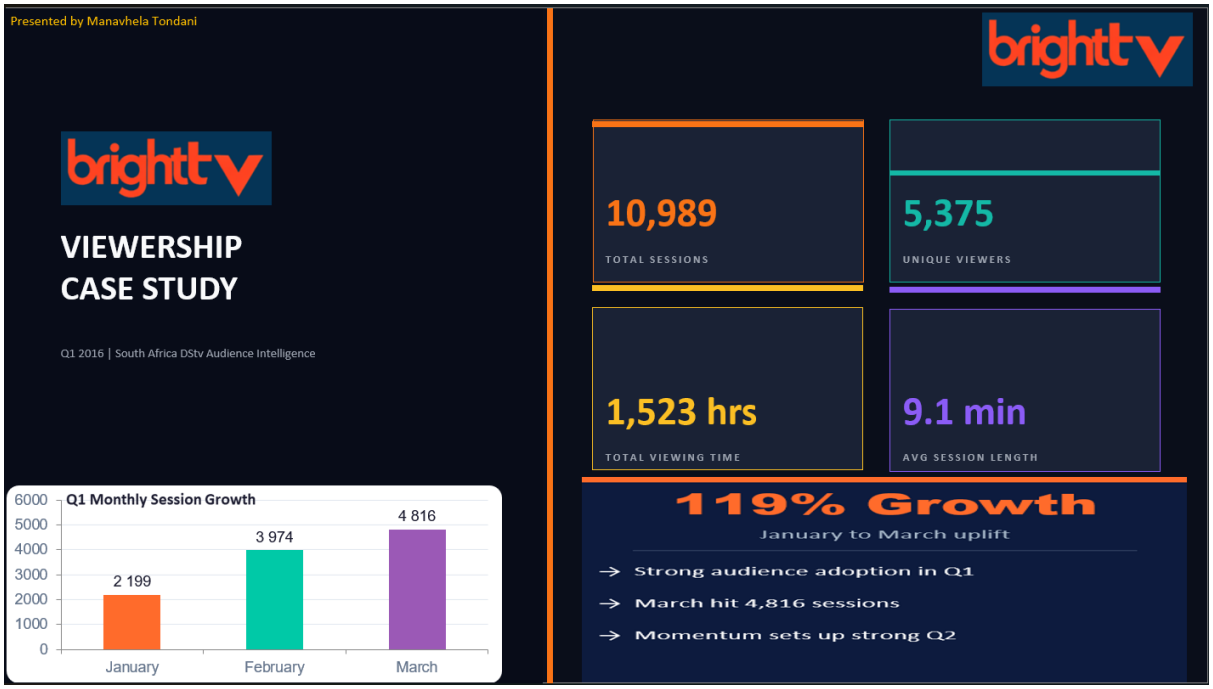
ameter

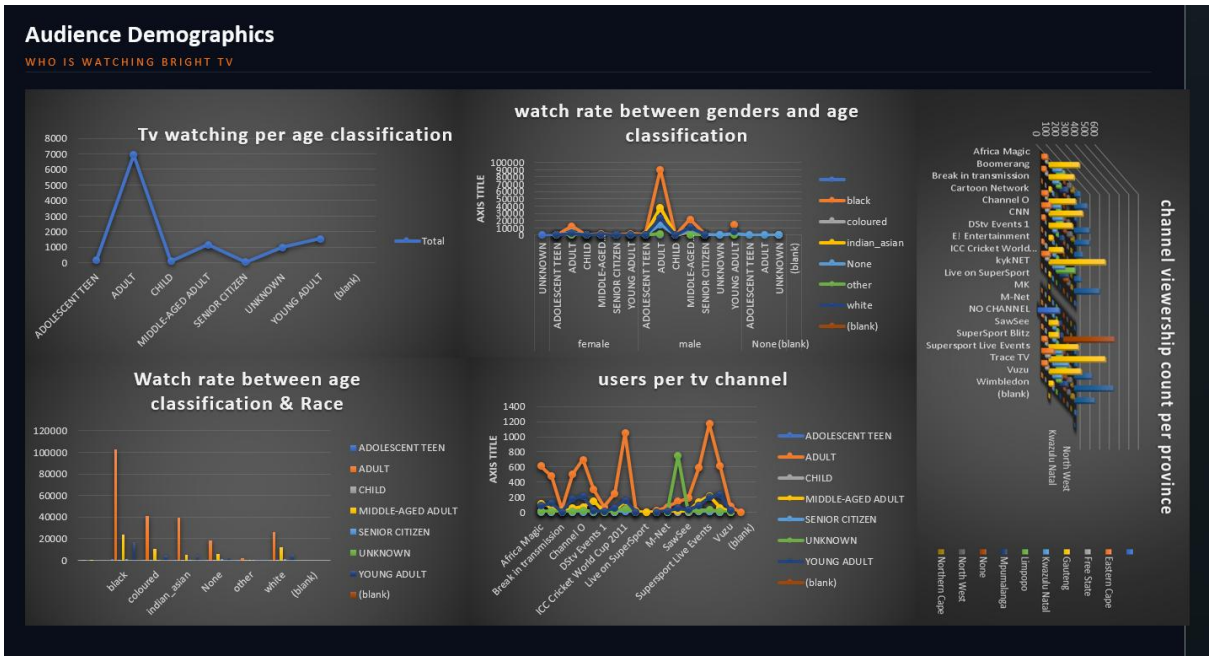
lts 10 of 10 → Table +

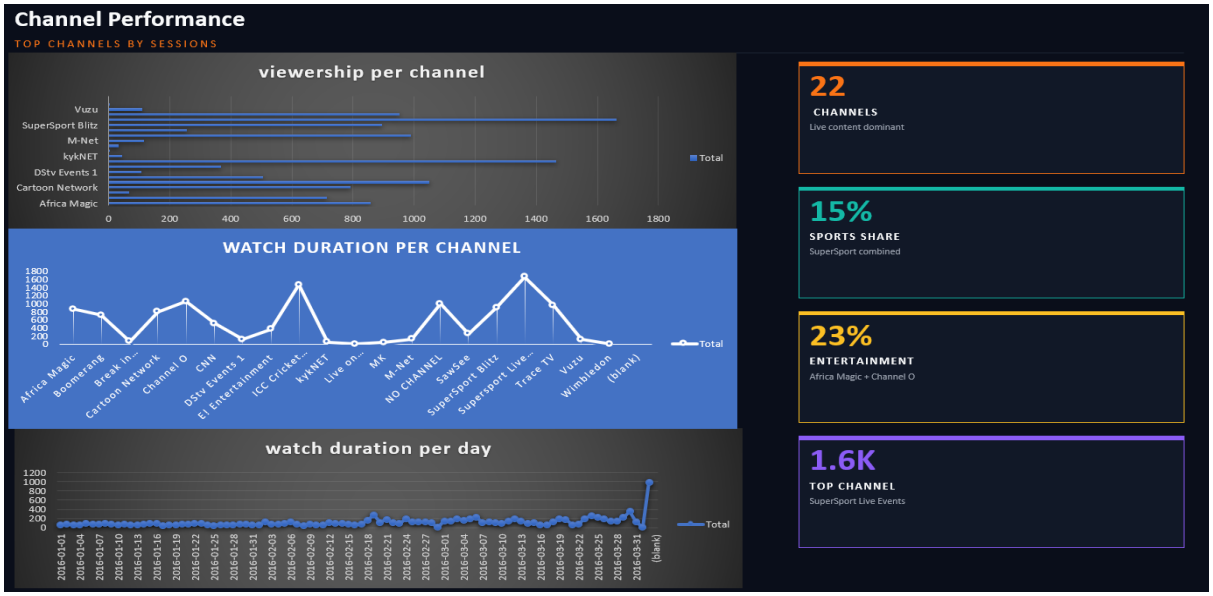
USER_ID	GENDER	AGE	AGE_CLASSIFICATION	RACE	PROVINCE	CHANNEL	RECORD_DATE	MONTH_NAME	RECORD_TI
33014	male	19	YOUNG ADULT	black	Limpopo	Boomerang	2016-04-01	Apr	01:24:00
33471	female	27	ADULT	None	Gauteng	Channel O	2016-04-01	Apr	00:08:00
93979	male	33	ADULT	indian_asian	Free State	Cartoon Network	2016-04-01	Apr	00:36:00
107967	male	27	ADULT	black	Gauteng	SuperSport Blitz	2016-04-01	Apr	00:04:00
789056	female	48	MIDDLE-AGED ADULT	black	Gauteng	CNN	2016-04-01	Apr	01:37:00
39624	male	19	YOUNG ADULT	black	Gauteng	Trace TV	2016-03-31	Mar	23:22:00
429528	male	30	ADULT	None	Gauteng	E! Entertainment	2016-03-31	Mar	23:47:00

3. Dashboard Creation

The result from The SQL Querry, was derived into an excel document to which dashboards were created and results are as follows:






Channel Performance
TOP CHANNELS BY SESSIONS
viewership per channel
Vuzu
SuperSport Blitz
M-Net
kykNET
DStv Events 1
Cartoon Network
Africa Magic
0 200 400 600 800 1000 1200 1400 1600 1800
Total
WATCH DURATION PER CHANNEL
1800 1600 1400 1200 1000 800 600 400 200 0
Africa Magic
Boomerang
Break In...
Cartoon Network
Channel O
CNN
DStv Events 1
E! Entertainment
ICC Cricket...
kykNET
Live on...
MK
M-Net
NO CHANNEL
SawSee
SuperSport Blitz
Supersport Live...
Trace TV
Vuzu
Wimbledon
(blank)
Total
watch duration per day
1200 1000 800 600 400 200 0
2016-01-01
2016-01-04
2016-01-07
2016-01-10
2016-01-13
2016-01-16
2016-01-19
2016-01-22
2016-01-25
2016-01-28
2016-01-31
2016-02-03
2016-02-06
2016-02-09
2016-02-12
2016-02-15
2016-02-18
2016-02-21
2016-02-24
2016-02-27
2016-03-01
2016-03-04
2016-03-07
2016-03-10
2016-03-13
2016-03-16
2016-03-19
2016-03-22
2016-03-25
2016-03-28
2016-03-31
(blank)
Total
22
CHANNELS
Live content dominant
15%
SPORTS SHARE
SuperSport combined
23%
ENTERTAINMENT
Africa Magic + Channel O
1.6K
TOP CHANNEL
SuperSport Live Events

Channel Performance
TOP CHANNELS BY SESSIONS
Watch duration per time slot
4000 3500 3000 2500 2000 1500 1000 500 0
AFTERNOON SLOT
EVENING SLOT
MIDNIGHT OWL
MORNING SLOT
(blank)
Total
Watch duration per day
(blank)
null
Sat
Fri
Thu
Wed
Tue
Mon
Sun
0 500 1000 1500 2000
Total
High viewership slot
Afternoon slot
3734 overall minutes
High viewership days
Friday
1675 hours
Count of USER_ID
TV WATCHING COUNT PER RACE
2%
1%
41%
15%
14%
15%
RACE
black
coloured
indian_asian
None
other
white
Count of USER_ID
TV WATCHER COUNT PER PROVINCE
2000 1500 1000 500 0
Eastern Cape
Free State
Gauteng
KwaZulu Natal
Limpopo
Mpumalanga
None
North West
Northern Cape
Western Cape
(blank)
RACE
black
coloured
indian_asian
None
other
PROVINCE
High viewership race
Gauteng
High TV watching Race
Black

Key Insights & Recommendations

STRATEGIC TAKEAWAYS FROM Q1 2016

01 Sports is the most watched

SuperSport Live Events and ICC Cricket World Cup together command 28% of all sessions, making live sports the single largest content category on the platform.

02 Male-Dominated Audience

81% of sessions are male-driven. Female engagement at just 9% presents a significant growth opportunity for targeted content acquisition and marketing.

03 Gauteng takes over viewership

Gauteng accounts for 34% of all sessions, followed by Western Cape at 17%. A Gauteng-first targeting strategy will capture the majority of high-value viewers.

04 Peak: Fri–Sat Evenings

Friday and Saturday are the top viewing days. Afternoon and evening slots combined account for 65% of sessions — prime scheduling windows for premium content.

05 Strong Q1 Growth

Sessions more than doubled from January (2,199) to March (4,816), an 119% uplift. This momentum signals strong audience adoption and retention heading into Q2.

06 Adult Core Audience

Adults (25–44) represent 63% of classified sessions. Combined with Young Adults, the 18–44 bracket is 77% — ideal for premium subscription and ad-targeting strategies.